## Questions

**Q1.**

The photograph shows a step-down transformer.

(a) Explain why step-down transformers are used in the transmission of electricity in the National Grid.

**(2)**

.....

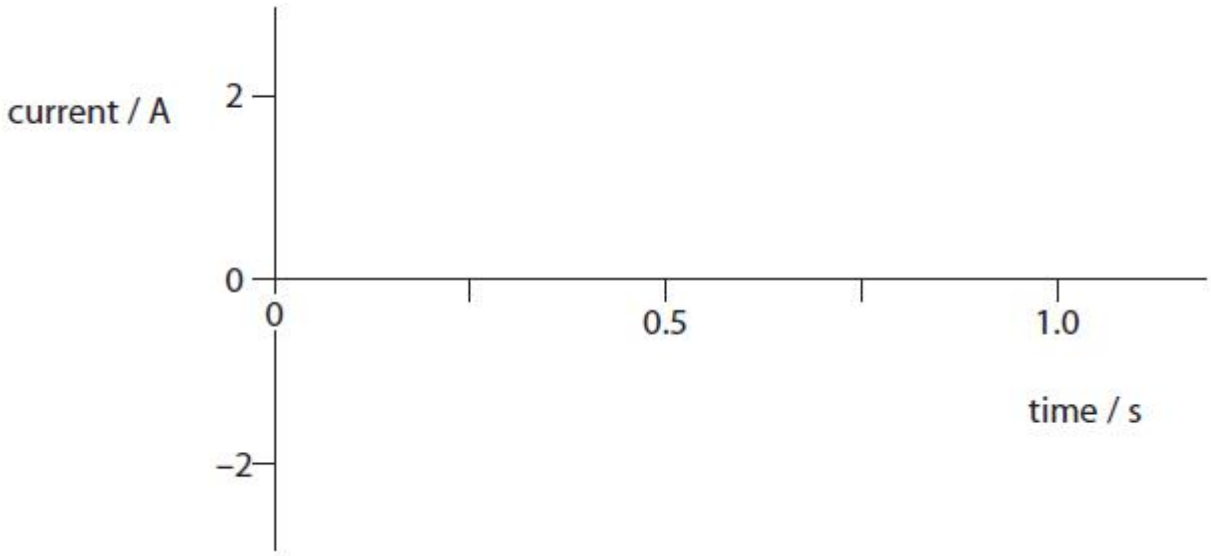
.....

.....

.....

(b) Transformers need alternating current to work properly.
Sketch a graph of an alternating current with a frequency of 2 Hz.

**(2)**




(c) A transformer has 2400 turns on the primary coil and 100 turns on the secondary coil.

Calculate the secondary voltage if the primary voltage is 12 V.

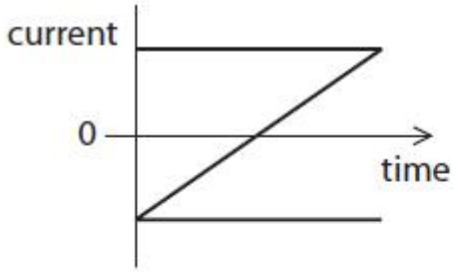
**(3)**

secondary voltage =..... V

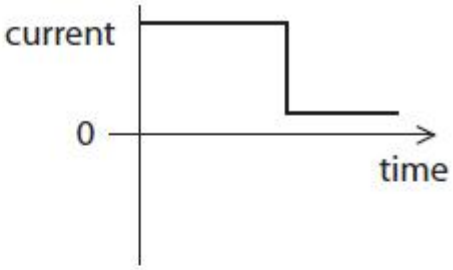
(d) Opening the switch in a circuit produces the opposite magnetic effect to closing the switch.
A scientist connected a switch, a fixed resistor and a battery to the primary coil of a step-up transformer.
The scientist also connected a fixed resistor across the secondary coil.
The scientist switched the circuit on and then later switched it off.

Which of these best represents the current in the secondary coil?

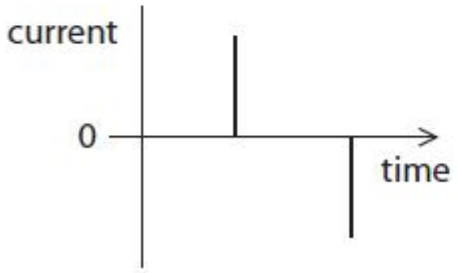
**(1)**



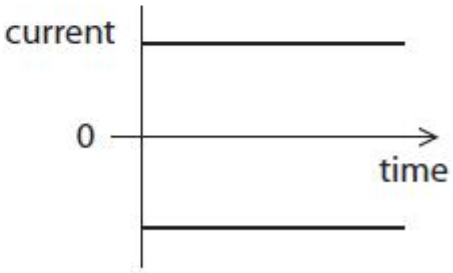

☐ **A**




☐ **B**




☐ **C**




☐ **D**

**(Total for Question = 8 marks)**

**Q2.**

* High voltage transmission cables and transformers are used in the national grid.

Explain how using high voltage transmission cables and transformers allows the distribution of electrical power around the United Kingdom to be as efficient as possible.

Refer to the following equations in your answer.

$$P = I^2 \times R$$

$$V_p \times I_p = V_s \times I_s$$

**(6)**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

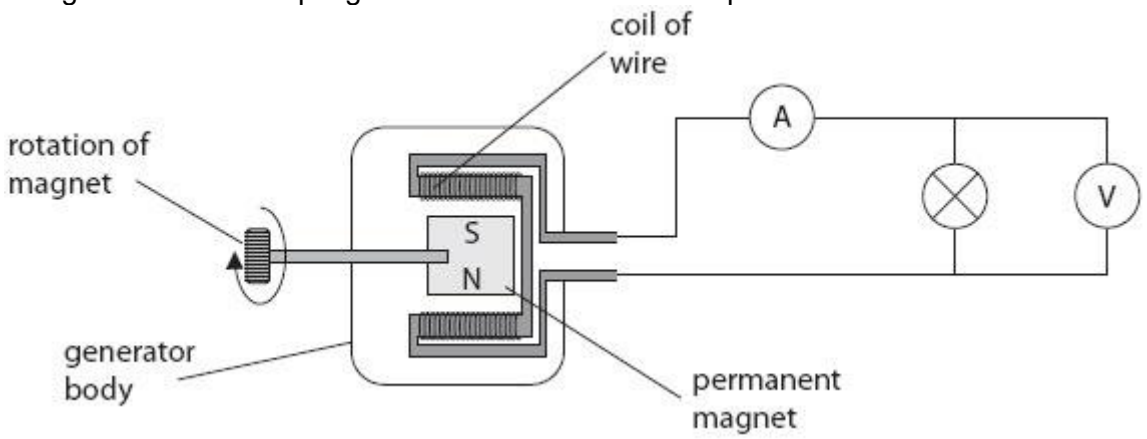
.....

.....

**(Total for question = 6 marks)**

**Q3.**

(a) The diagram shows a simple generator connected to a lamp.




The magnet is made to spin at a steady speed.
The ammeter gives a reading of 1.5 A.
The voltmeter gives a reading of 6 V.

(i) Calculate the output power of the generator.

**(2)**

.....

(ii) State two changes to the design of the generator that would give a larger output power for the same speed of rotation.

**(2)**

.....

.....

(iii) This generator supplies an alternating current (AC) to the lamp.
Other types of generators supply a direct current (DC).
Describe the difference between charge movement in a direct current and in an alternating current.

**(2)**

.....

.....

*(b) The first public power station was built in the centre of New York.

It used generators to supply direct current at 110 V. The cables had to go underground and they could only supply
nearby shops and offices.

The electricity was mainly used for electric light.

The development of alternating current generators led to major changes in the way electricity is transmitted and used.

Compare the modern National Grid system with the early system in New York.

**(6)**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

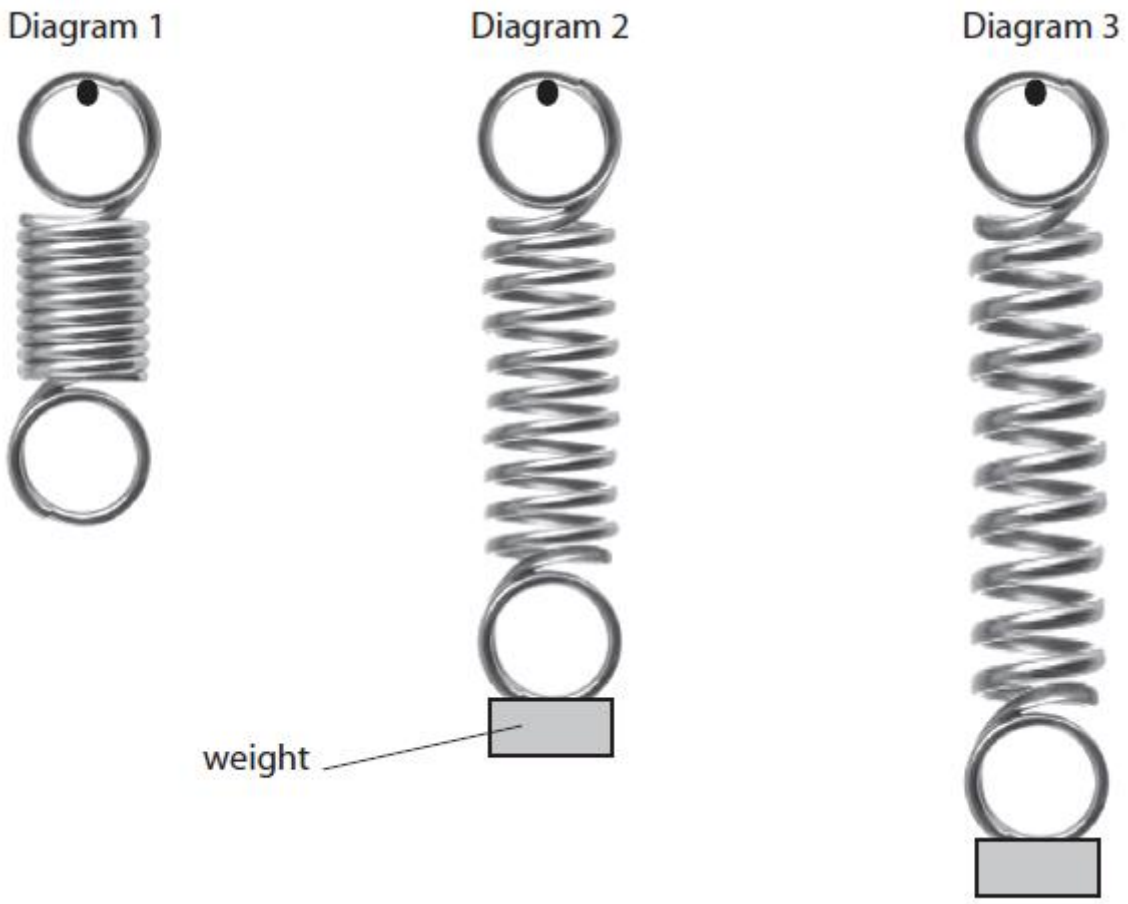
.....

**(Total for Question is 12 marks)**

**Q4.**

(a) The diagrams show a spring hanging from a nail.

- In Diagram 1 there is no weight on the spring.
- Diagram 2 shows the spring after a weight is added.
- Diagram 3 shows the spring after the weight has been pulled down slightly.




(i) Complete the sentence by putting a cross (☒) in the box next to your answer.
When held stationary as in Diagram 3,

**(1)**

☐ **A** the spring has zero elastic potential energy
☐ **B** the weight has equal amounts of elastic potential and kinetic energy
☐ **C** the weight has more kinetic energy than gravitational potential energy
☐ **D** the spring has more elastic potential energy than the weight has kinetic energy

(ii) The spring is stretched from the position shown in Diagram 2 to the position shown in Diagram 3.
The spring is then released.
Describe the energy changes that take place until the spring stops vibrating.

**(3)**

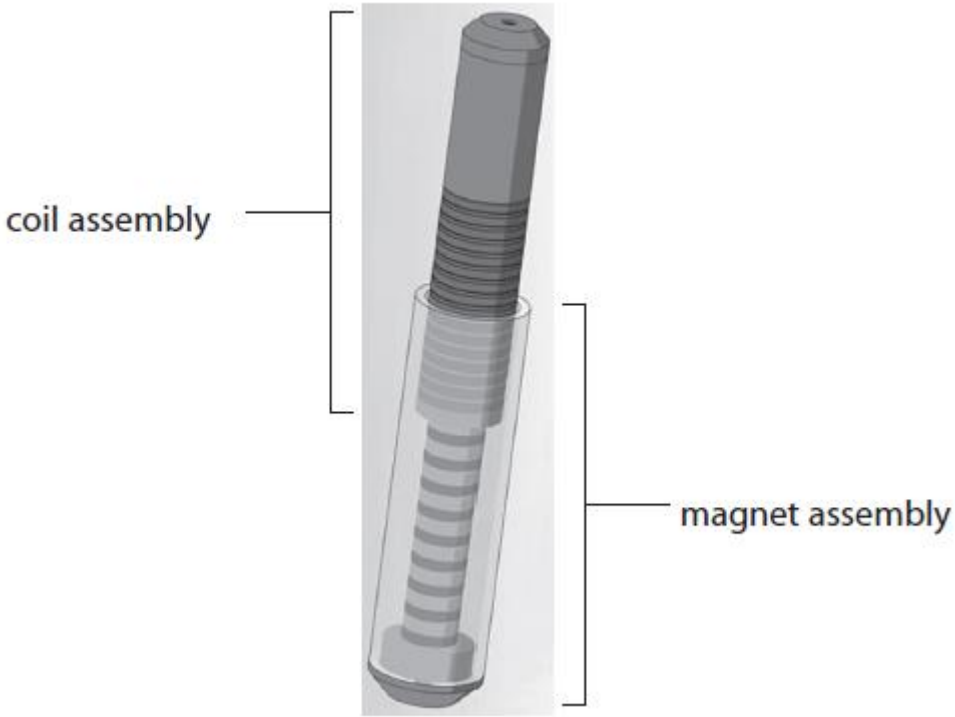
.....

.....

.....

.....

(b) Shock absorbers with springs are used on some motorcycles.
These shock absorbers reduce the bounce on an uneven road.
A new shock absorber has been developed to convert some of the movement energy into another form.
It consists of magnets which slide inside a coil when the motorcycle goes over a bump.




Some of the energy which would otherwise be wasted can be recovered and so fuel is saved.
(i) Complete the sentence by putting a cross (☒) in the box next to your answer.

**(1)**

☐ **A** increase the thermal energy obtained from the fuel
☐ **B** increase the efficiency of the motorcycle
☐ **C** decrease the speed of the motorcycle
☐ **D** decrease the braking power of the motorcycle

(ii) Explain how this new type of shock absorber can provide electrical energy.

**(2)**

.....

.....

.....

.....

(iii) The diagram shows the bumps on the surface of two roads, L and M.
why the device will transfer more energy on road L than on road M for a motorcycle travelling at the same speed.

**(3)**

road L

road M

.....

.....

.....

.....

.....

.....

**(Total for Question = 10 marks)**

**Q5.**

A teacher is demonstrating electromagnetic induction.
The teacher has a bar magnet, a coil of wire and a sensitive voltmeter.

(i) Draw a diagram to show how the teacher should arrange the apparatus.

**(1)**

(ii) Explain how the teacher could use this apparatus to demonstrate the factors affecting the size and direction of the induced potential difference.

**(4)**

.....

.....

.....

.....

.....

.....

.....

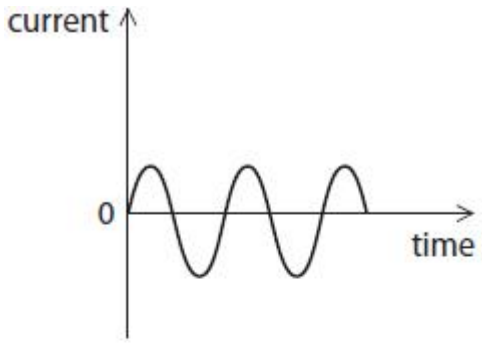
.....

**(Total for question = 5 marks)**

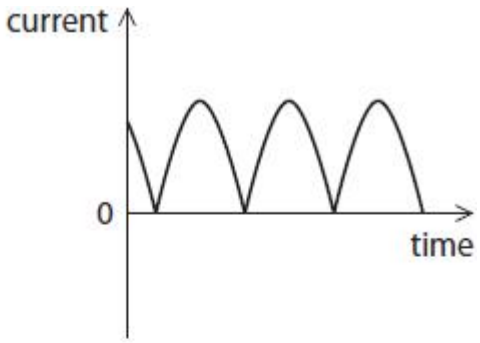
**Q6.**

Which of these could be the output for a dynamo?

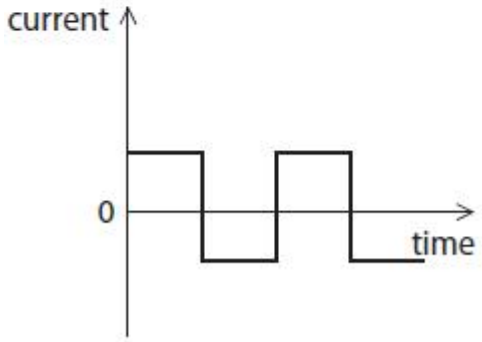
(1)



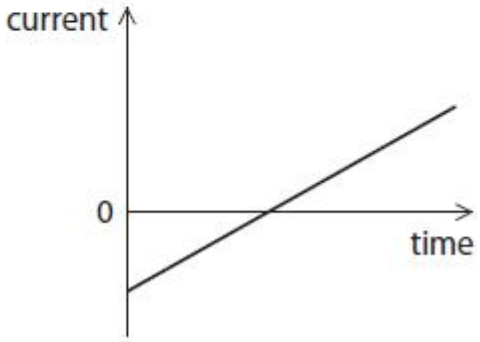

☒ **A**




☒ **B**




☒ **C**




☒ **D**

**(Total for question = 1 mark)**

**Q7.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Figure 15 shows three stages of a magnet moving into and then out of a coil of wire.

The coil is connected to a milliammeter.

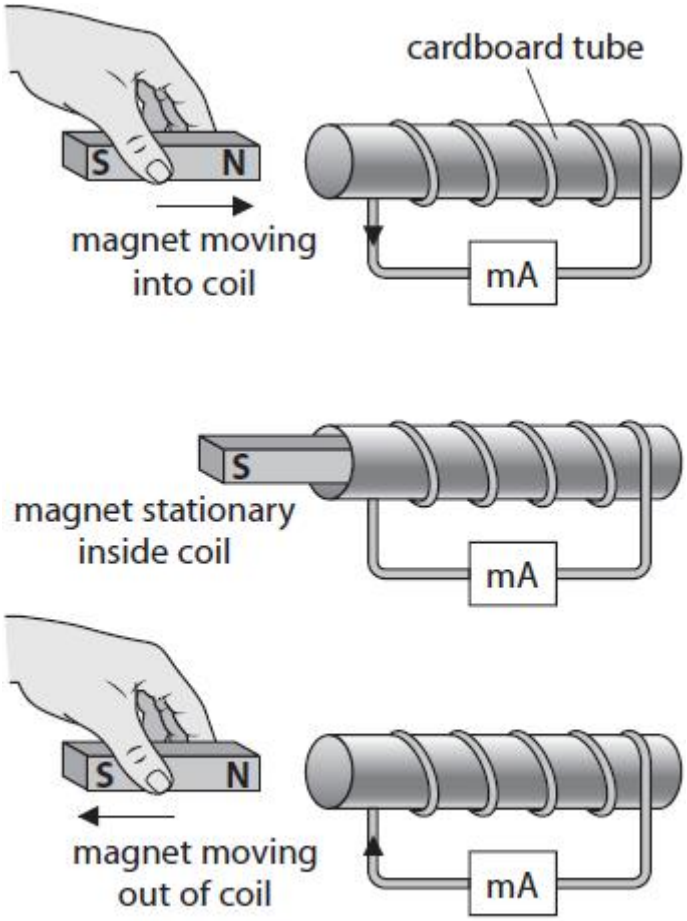



**Figure 15**

(i) Which row of the table shows the deflection on the milliammeter for the three stages in Figure 15?

**(1)**

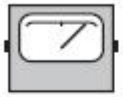
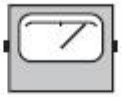
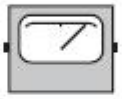
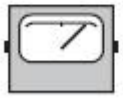
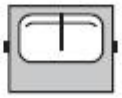
| | magnet moving into coil | magnet stationary inside coil | magnet moving out of coil |
|---|---|---|---|
| ☐ A |  |  | |
| ☐ B | |  | |
| ☐ C | | | |
| ☐ D |  |  | |

**Figure 16**

(ii) Give **two** ways of increasing the deflections on the milliammeter shown in Figure 16.

**(2)**

1 .....

.....

2 .....

.....

**(Total for question = 3 marks)**

**Q8.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Figure 23 shows a model dynamo.

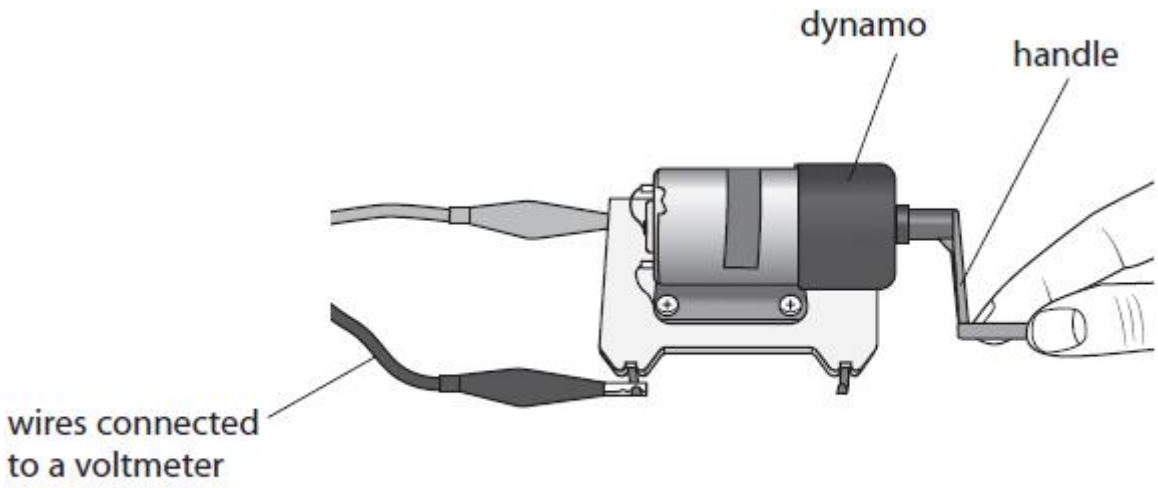



**Figure 23**

The dynamo contains a coil of wire that can spin inside a permanent magnet.

The dynamo produces a D.C. output.

A teacher connects a voltmeter to the terminals of the dynamo.

The teacher rotates the handle to make the coil spin inside the magnet.

Figure 24(a) shows the reading on the voltmeter.

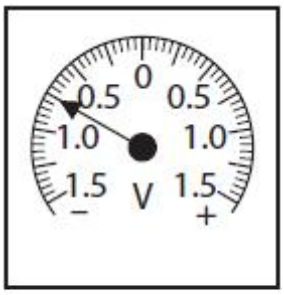



**Figure 24(a)**

The teacher then rotates the handle differently.

Figure 24(b) shows the new reading on the voltmeter.

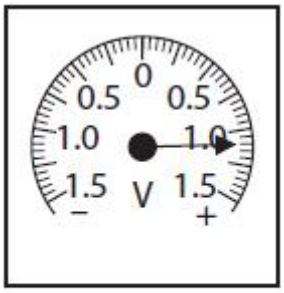



**Figure 24(b)**

(i) Which row of the table shows how the rotation of the handle has changed between (a) and (b)?

**(1)**

| | speed of rotation | direction of rotation |
|---|---|---|
| ☐ A | (a) faster than (b) | (a) opposite to (b) |
| ☐ B | (a) faster than (b) | (a) same as (b) |
| ☐ C | (a) slower than (b) | (a) opposite to (b) |
| ☐ D | (a) slower than (b) | (a) same as (b) |

(ii) The teacher connects the dynamo to a lamp.

It is now more difficult for the teacher to rotate the handle.
Explain why it is more difficult to turn the dynamo when it is connected to a lamp.

**(2)**

.....

.....

.....

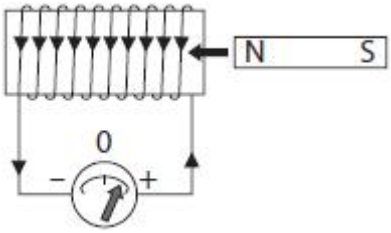
.....

**(Total for question = 3 marks)**

**Q9.**

Figure 11 shows a magnet moving into a coil.

This generates a current.




The magnet is now dropped through three identical coils as shown.

The coils are connected to a data logger that can produce the graph shown in Figure 12.

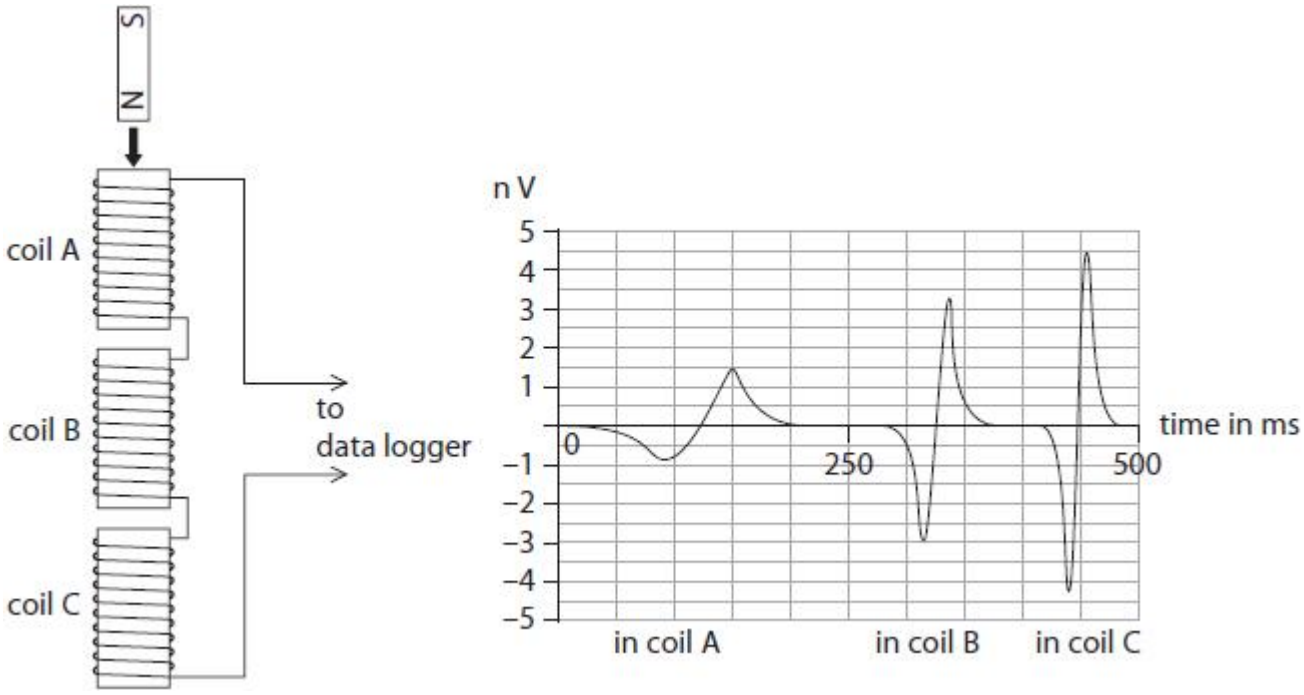



**Figure 12**

(i) State and explain **two** ways in which the voltage-time graph shows you that the magnet accelerated as it travelled through the coils.

**(4)**

1 .....

.....

.....

.....

2 .....

.....

.....

.....

(ii) Explain why the meter in Figure 11 could not be used to obtain the data for the graph in Figure 12.

**(2)**

.....

.....

.....

.....

.....

**(Total for question = 6 marks)**

## Mark Scheme

Q1.

| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(a)** | An explanation linking two from | | |
| | MP1 (so that they) decrease the (high) voltages (1) | **stepping down voltage reducing from {high/eg 200 000 V} to {low /e.g.230 V} voltage** | |
| | MP2 high voltages used for efficiency/energy saving (1) | **low current used for efficiency/ energy saving** | |
| | MP3 (step-down transformers) used {near / for} {homes / factories/appliances} (1) | | |
| | MP4 (so that it is) safer (1) | less risk of electrocution<br>high voltages are dangerous | **(2)** |

| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(b)** | one line / curve above **and** below x-axis (1) | | |
| | two complete cycles in the 1.0 s (1) | one complete cycle in 0.5 s | **(2)** |

| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(c)** | Transposition (1)<br><br>$V_s = V_p \times n_s/n_p$<br><br>Substitution (1)<br><br>$(V_s =) \frac{12 \times 100}{2400}$<br><br>Evaluation (1)<br><br>0.5 (V) | <br><br>Substitution and transposition in either order<br><br><br>i.e. if $\frac{12 \times 100}{2400}$ is seen this scores 2<br>If they sub Vp, Np and Ns correctly, ignore anything for Vs even a blank<br><br>Calculation may be done using turns ratio<br>Correct answer no working = full marks<br>answer (no working) with POT error =2 (eg 5 or 0.05)<br>Ignore powers of 10 until evaluation | **(3)** |

| **Question Number** | Answer | Acceptable answers | **Mark** |
|---|---|---|---|
| **(d)** | C | | **(1)** |

**Q2.**

| Question Number | Answer | Mark |
|---|---|---|
| * | Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme.<br>The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant.<br><br>AO1(6 marks)<br>Understanding of physics<br>• (long) transmission wires have resistance<br>• reduced p.d. at the destination<br>• (thermal) energy is dissipated in the transmission wires<br>• this creates a power loss (refers to $P=I^2R$)<br>• transformers are used to step up to a high voltage for transmission<br>• this means a low current (refers to $V_PI_P=V_SI_S$)<br>• so power loss is small(er)<br>• transformers used to step down to a safer voltage for consumers<br>• consumer wires are shorter and so power loss is less of an issue | **(6)**<br>AO 1 1 |

| Level | Mark | Descriptor |
|---|---|---|
| | 0 | • No rewardable material. |
| Level 1 | 1-2 | • An explanation that demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1)<br>• Presents an explanation that is not logically ordered and with significant gaps. (AO1) |
| Level 2 | 3-4 | • An explanation that demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1)<br>• Presents an explanation of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1) |
| Level 3 | 5-6 | • An explanation that demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1)<br>• Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1) |

**Q3.**

| | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(ai)** | Substitution (1)<br>1.5 × 6<br>Evaluation (1)<br>9 (W) Ignore any unit given by candidate. | Power of 10 error max 1 mark Give full marks for correct answer with no working shown | **(2)** |
| **(aii)** | • More turns on the coil (1)<br>• More powerful/stronger magnet(s) (1) | Wrap coils on iron (core/former)/ more coils/twists/loops. Bigger coil is insufficient. More magnets. Bigger/larger magnet is insufficient. Ignore increase speed of rotation | **(2)** |
| **(aiii)** | A description including<br>• in one direction | 'DC goes straight' is insufficient AC switches/changes | **(2)** |

| | | | |
|---|---|---|---|
| | only for DC (1)<br>• reversing direction for AC (1) | direction OR moves to and fro 'AC goes different ways' is insufficient.<br>Diagram with labelled arrows could get 2 marks. | |

| | | Indicative Content | Mark |
|---|---|---|---|
| **QWC** | ***(b)** | A comparison including some of the following ideas<br>• Transformers can be used or voltages/currents can be changed/transformed<br>• AC (can transmit) at lower current/high(er) voltage<br>• National Grid is (usually) over ground (DC cables (were) underground)<br>• Less energy lost in transmission<br>• National Grid system can supply to customers further away<br>• Possible to create a grid linking power stations<br>• More flexibility in voltage for consumer<br>• Consumer can draw large(r) current<br>• More flexibility in power drawn<br>• Great(er) range of devices can be powered Ignore methods of electricity production | **(6)**<br>**Exp** |
| **Level** | **0** | No rewardable content | |
| **1** | **1 - 2** | • a limited (maybe implied) comparison giving one fact e.g: AC can be at high(er) voltage OR the National Grid can supply houses not close to a power station/ further (away/than the New York system.)<br>• the answer communicates ideas using simple language and uses limited scientific terminology | |

| | | |
|---|---|---|
| | | • spelling, punctuation and grammar are used with limited accuracy |
| **2** | **3 - 4** | • a simple comparison including two ideas which may be linked or not eg Nat. Grid can supply whole country and can be used for more appliances (than just lighting). e.g: AC can be transmitted further (than DC) (because it) wastes less energy<br>• the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately<br>• spelling, punctuation and grammar are used with some accuracy |
| **3** | **5 - 6** | • A detailed comparison including at least three ideas, with at least one direct link between two of them.<br>• e.g. AC can be transmitted further (than DC) because AC can be transformed to lower current/high(er) voltages.<br>OR<br>AC can be transformed to lower current/high(er) voltages.<br>Greater range of devices used.<br>• the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately<br>• spelling, punctuation and grammar are used with few errors |

Total for Question = 12 mark

**Q4.**

| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(a)(i)** | D the spring has more elastic potential energy than the weight has kinetic energy | | **(1)** |

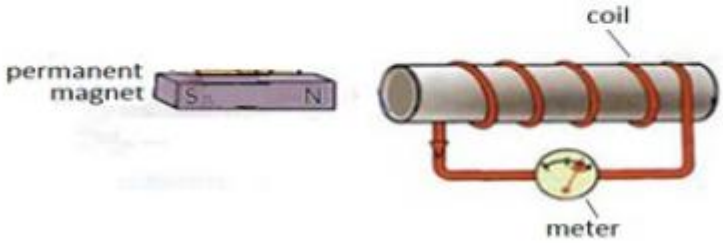
| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(a)(ii)** | A description including three from<br><br>MP1 Elastic potential energy /EPE (in stretched spring) (1)<br><br>MP2 (EPE is) transferred to KE (initially) (1)<br><br>MP3 change from KE to GPE or vice versa(1)<br><br>MP4 (correct idea of) energy changes continuing<br><br>MP5 {total mechanical energy /kinetic +potential energy} decreases (continuously) (1)<br><br>MP6 (Eventually all is transferred to) {thermal/heat} (energy) (1) | care should be taken not to award marks for contradictory examples<br>Starting point for description does not matter<br>Ignore sound energy<br><br>EPE becomes/goes to KE (initially)<br><br>condone<br>amplitude decreases to zero<br>KE or PE 'lost' to surroundings | **(3)** |

| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(b)(i)** | B increase the efficiency of the motorcycle | | **(1)** |

| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(b)(ii)** | MP1 (bump produces) relative motion (1)<br><br>MP2 (motion between magnet and coil) {induces / generates} voltage (1) | coil moves round magnet/magnet moves {into/out of} coil / coil {cuts / moves across} magnetic field<br>ignore magnets slide inside a coil (see stem)<br><br>electromagnetic induction<br>condone {induces / generates }<br><br>{current/electricity}<br><br>ignore (see stem)<br>electrical energy<br>provides / produces | **(2)** |

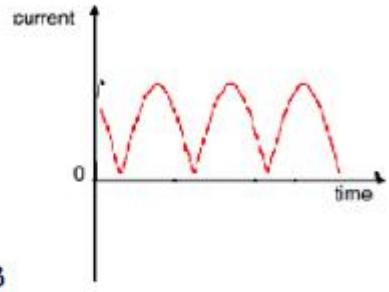
| Question Number | Answer | Acceptable answers | Mark |
|---|---|---|---|
| **(b)(iii)** | An explanation linking<br><br>MP1 {more/frequent} bumps (1) (idea of shorter time / increased frequency)<br><br>MP2 (bigger bumps produce) bigger amplitude / move more up and down (idea of bigger size) (1)<br><br>MP3 (so) {induced voltage /voltage generated} is larger (1) | <br><br>idea of up and down for bump (coil / magnets) move up and down {faster / more often}<br><br>(coil/magnets) move {further/higher/bigger distance} (up and down)<br><br>{induced current/current generated} is larger<br>electromagnetic induction gives more voltage/current<br><br>condone more electricity/electrical energy is {induced / generated}<br><br>allow once for MP1 (if MP1 or MP2 is not scored):<br>'bumpier'<br>'go in and out more' | **(3)** |

**Q5.**

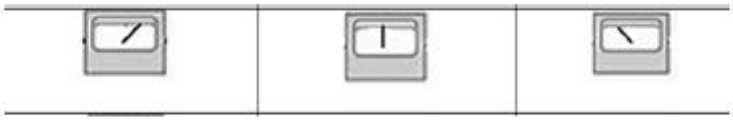
| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | a diagram that has the meter connected across the ends of a coil and a magnet orientated parallel to the axis of the coil; for example   | poles need not be labelled | (1) |

| Question Number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | An explanation linking<br><br>move magnet towards coil and then away from coil (1)<br><br>with<br>note change in 'direction' of **meter** (1)<br><br>move magnet quickly then slowly (1)<br><br>with<br><br>larger / smaller **meter** reading (1) | <br><br>change poles of magnet<br><br>allow use of ± in digital meters<br><br>change speed of movement of magnet<br>or<br>changes to the number of turns<br><br>ignore changes to size/strength of magnet | (4) |

**Q6.**

| Question Number: | Answer | Mark |
|---|---|---|
| |  <br>B<br><br>**The only correct answer is B**<br><br>***A*** *is incorrect because it shows an alternating current which is produced by an alternator and not by a dynamo*<br>***C*** *is incorrect because it shows a square waveform which is not produced by a dynamo*<br>***D*** *is incorrect because it shows current linearly increasing with time and this is not produced by a dynamo* | **(1)**<br>AO 3 2b |

**Q7.**

| Question number | Answer | Mark |
|---|---|---|
| **(i)** | C<br>  <br><br>A and B are incorrect because there is no current when the magnet is station in the coil.<br>D is incorrect because there is always a current when the magnet is moving in th coil | **(1)** |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(ii)** | any two from<br><br>moving the magnet faster (1)<br><br>using a stronger magnet (1)<br><br>more turns/rotations on the coil (1) | <br><br><br><br><br><br>do not allow increase size of coil | **(2)** |

**Q8.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| **(i)** | C (a) slower than (b) and (a) opposite to (b)<br>B and D are incorrect because the direction of the readings are opposite and so the rotations must be opposite<br>A is incorrect because the value in (a) is less than the value in (b) and so the rotation cannot be faster | | **(1)**<br>**AO1** |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | **either**<br><br>an answer in terms of magnetic fields and forces:<br><br>current in the (dynamo) coil generates a magnetic field (1)<br><br>(interaction of) fields cause force (that opposes motion) (1)<br><br>**OR**<br><br>an answer in terms of output energy and input work:<br><br>energy is being transferred to the lamp (1)<br><br>additional work is being done (by teacher) / additional energy is necessary (1) | accept lamp uses (thermal / heat / light) energy (to light up)<br><br>ignore reference to greater (electrical) resistance | **(2)**<br>**AO1** |

**Q9.**

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (i) | An explanation that combines identification via a judgment to reach a conclusion, via reasoning to include<br>• the peaks get higher (1)<br>• because (faster means) greater rate of change of magnetic field (1)<br><br>• the width of the peaks gets less (1)<br>• because (faster means) shorter times for magnet to travel through (coils) (1) | <br><br><br>greater induced emf / voltage<br><br><br>width of each wave gets less | (4) |

| Question number | Answer | Additional guidance | Mark |
|---|---|---|---|
| (ii) | An explanation that combines identification, via a judgment, to reach a conclusion, via reasoning, linking one from:<br><br>• time involved is very short (1)<br><br>• 500 ms (0.5 s) shown on graph (1)<br><br>with one from<br><br>• the meter could not respond quickly enough (1)<br><br>• human/person could not take/record the readings quickly enough (1) | <br><br><br><br><br><br><br><br>data loggers can take (lots of) readings quickly<br><br>human reaction times insufficient | (2) |